

To: Marin County Planning Department and Board of Supervisors

From: Sarayu Ramnath

Date: November 16th, 2025

Subject: Scaling the Bolinas Pilot Septic Program to Strengthen County Coastal Resilience

Executive Summary

This memo recommends that County of Marin expands the Bolinas Pilot Septic Program into a permanent decentralized wastewater program to strengthen coastal resilience and reduce contamination risk. Bolinas Mesa, a rural neighborhood in Marin County, CA, faces coastal flooding, erosion, and rising groundwater levels that threaten aging septic systems. In this high-groundwater coastal zone, failing systems can release untreated wastewater into the local ecosystem and recreation areas, compromising public health and ecological habitat. The existing Bolinas Pilot Septic Program, which provides limited technical and financial support for upgrading high-risk or unpermitted systems, offers a foundation to address these risks. Expanding the program to cover all high-risk properties on the Mesa would reduce vulnerability and protect the coastal environment.

Problem and Background

Bolinas Mesa, in the unincorporated town of Bolinas in Marin County hosts aging septic systems in a high-groundwater zone ⁸. While Bolinas has a centralized wastewater system, it mainly serves the downtown area, and a moratorium on new sewer connections combined with limited infrastructure capacity ² means that Mesa residents rely on on-site septic systems. These systems have previously failed after high-impact weather events, causing beach closures and releasing untreated wastewater downstream ³. The Bolinas Community Public Utility District recently established a special category of Septic System Limited Water Use Permits to support the pilot program ¹, demonstrating local regulatory support. This is an intersecting conservation and public health issue that will be exacerbated if adaptations are not made, highlighting the need for proactive community response and coordinated county support.

Key Findings

1. **Risks to Health, Infrastructure & Habitat:** Bolinas Mesa faces increasing risks from sea level rise, king tides, and storm events, which raise groundwater levels and increase the likelihood of on-site septic system failures ⁸. Past leaks have caused beach closures and water quality advisories, showing direct public health impacts ⁵. Because downtown Bolinas sits downhill, septic failures on the Mesa could disrupt critical infrastructure such as the post office, grocery store, gas station, and library. These

failures also threaten downstream coastal ecosystems such as Bolinas Lagoon, local beaches, and Duxbury Reef State Marine Conservation Area.

2. **Community Equity:** Many Mesa residents rely on costly individual systems, disproportionately affecting lower-income households. About one-third of households in Bolinas earn below \$50,000, 12.8% of residents live below the poverty line, and 25% of households are renter-occupied ¹¹, underscoring the importance of accessible support for vulnerable residents.
3. **Policy & Funding Frameworks:** The Bolinas Pilot Septic Program, led by the Bolinas Community Land Trust, operates with strong local support and partnerships with Marin County permitting authorities, including Bolinas Community Public Utility District (BCPUD) and the County Community Development Agency. The County of Marin has already provided \$320,000 in waived permit and road impact fees to support the pilot program, building on earlier efforts to ease regulatory requirements for unpermitted septic systems ⁷. In June 2025, BCPUD passed Resolution 743, establishing a special category of “Septic System Limited Water Use Permits” for the pilot program ¹.
4. **Relevant Programs:** Coastal and rural regions in California are beginning to address septic vulnerability under sea level rise and groundwater changes. Santa Cruz County’s Local Coastal Program allows exceptions for septic upgrades in rural areas when failing systems pose public health risks, aligning with the goals of the pilot program ⁴. Within Marin County, Stinson Beach Water District and Marin Environmental Health Services are conducting a wastewater feasibility study to explore adaptive approaches ⁶. Regionally, the North Coast Resource Partnership has initiated decentralized wastewater planning for rural communities, showing that structured, risk-based programs are becoming more common ⁹.

Recommendations

1. **Expand Program from Pilot to Permanent:** To address the growing risks from aging septic systems and climate-driven impacts, the Bolinas Pilot Septic Program should be scaled up to reach all high-risk properties on the Mesa. Expansion should include on-site upgrades and clustered community systems where feasible, guided by a Mesa-wide, risk-prioritized assessment to identify remaining high-risk systems and evaluate cost-effective options, building on the framework of BCPUD Resolution 743.
2. **Align Regulatory Agencies:** County of Marin should formalize coordination among Bolinas Community Land Trust, Bolinas Community Public Utility District and the county Community Development Agency and Environmental Health Services to streamline permitting, align standards, and provide a clear governance structure.

Bolinas Community Land Trust should continue to lead program oversight and participant coordination.

3. **Develop a Community Advisory Board:** Establish a Mesa Neighborhood Advisory Board to serve as community representatives to Bolinas Community Land Trust, guiding program design, sharing information with residents, and providing feedback on program scaling and implementation. The Advisory Board will ensure transparency, equity in upgrade prioritization, and broad community participation, which will build trust as the program expands.
4. **Leverage State and Federal Funding:** The County of Marin should secure and allocate state and federal funds to support program expansion, including U.S. Department of Agriculture Waste & Water Disposal Loan and Grant Program and the Environmental Protection Agency Clean Water State Revolving Fund. These funds can support system upgrades and repairs, new or clustered system construction, and maintenance infrastructure such as septage treatment works^{10,12}. County-backed applications for these programs will enable scaling the pilot to a comprehensive Mesa-wide program.

Conclusion

Together, these recommendations provide a feasible pathway for County of Marin to expand BCLT's septic upgrade program from a limited pilot into an established county-aligned initiative. With sustained funding, community engagement, and coordinated regulatory support, the program can reach more unserved households. Bolinas is not the only rural coastal community facing this challenge, and the pilot program is an innovative, community-driven approach to climate-adaptive infrastructure. With targeted County support, the program can strengthen public health protections, safeguard local ecosystems, and provide a replicable framework for rural coastal resilience efforts statewide.

References

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